Problem Ladders v0.3

2026-10-09 · 20 ladders · 112 problems · internal review copy, not for students

Part 1 has the problems only. Try them before looking at Part 2 (answer key) and Part 3 (solutions, idea cards and how each answer was checked by computer).

Ladder	Name	Steps	Idea card (internal)
I	Three Divisors Make a Total	4 + Challenge	SL2022-N1
J	Root or Add Three	5 + Challenge	SL2017-N1
K	Divisor Counts That Double	5 + Challenge	SL2011-N1
L	The Shrinking Product	5 + Challenge	SL2010-N1
M	The Average Divisor Count	5 + Challenge	SL2006-N3
N	Values That Share a Factor	5 + Challenge	SL2016-N3
O	Pick With Small Gaps	5 + Challenge	SL2022-C1
P	Rainbow Necklaces	4 + Challenge	SL2021-C2
Q	Flip the k-th Coin	5 + Challenge	SL2019-C3
R	Rooks and Empty Squares	5 + Challenge	SL2014-C3
S	Pairs With Different Sums	5 + Challenge	SL2012-C2
T	Triples Without Repeats	5 + Challenge	SL2009-C2
U	Doubling Prefix Sums	4 + Challenge	SL2008-C2
V	Same-Colour Triples	5 + Challenge	SL2007-C3
W	Lattice Friends	4 + Challenge	SL2008-C3
X	Floors of Quotients	5 + Challenge	SL2021-A3
Y	Sums That Divide the Total	4 + Challenge	SL2011-A1
Z	Average of the Last Few	4 + Challenge	SL2018-A4
AA	The Inner Arc	4 + Challenge	SL2006-G1
AB	Powers Plus Fourth Powers	4 + Challenge	SL2023-N2

Part 1 — Problems

Ladder I — Three Divisors Make a Total

A divisor of N is a positive whole number that divides N exactly (1 and N count). The three divisors in each problem must be different.

Step 1 Kangaroo	The divisors of 12 are 1, 2, 3, 4, 6, 12. In how many ways can you choose three different divisors of 12 that add up to 19?
Step 2 Kangaroo / MATHCOUNTS	What is the smallest whole number N such that three different divisors of N add up to N itself?
Step 3 MATHCOUNTS	Three different divisors of N are N/a, N/b and N/c, with $a < b < c$, and they add up to N. What is $a + b + c$?
Step 4 AMC 8	What is the smallest whole number N that has three different divisors adding up to 100?
Challenge **AIME**	What is the smallest whole number N that has three different divisors adding up to 2026?

Ladder J — Root or Add Three

Start with a whole number. At each step: if the number is a perfect square, replace it by its square root; otherwise add 3. Write down every number you get.

Step 1 Kangaroo	Start at 7. The first number written is 7. What is the 6th number written?
Step 2 Kangaroo / MATHCOUNTS	Start at 5. How many of the first 100 numbers written are perfect squares?
Step 3 MATHCOUNTS	When a perfect square is divided by 3, how many different remainders are possible?
Step 4 AMC 8	Start at 2026. What is the smallest number that is ever written?
Step 5 AMC 10	Start at 30. After a while, the same few numbers repeat forever. What is the sum of those repeating numbers?
Challenge **AIME**	For how many starting numbers from 1 to 1000 is some number written twice?

Ladder K — Divisor Counts That Double

For a whole number n, d(n) is the number of its divisors. A **block** is a number of the form p, p^2, p^4, p^8, ... where p is a prime (the exponent is a power of 2).

Step 1 Kangaroo	How many divisors does 24 have?
Step 2 Kangaroo / MATHCOUNTS	What is the smallest number with exactly 4 divisors?
Step 3 MATHCOUNTS	What is the smallest number with exactly 8 divisors?
Step 4 AMC 8	How many blocks are smaller than 30?
Step 5 AMC 10	What is the smallest number with exactly 64 divisors?
Challenge AIME	Let N be the smallest number with exactly 1024 divisors. How many different primes divide N?

Ladder L — The Shrinking Product

Choose different whole numbers s, each at least 2, and multiply the fractions $(1 - 1/s)$ together.

Step 1 Kangaroo	What is $(1 - 1/2)(1 - 1/3)(1 - 1/4)(1 - 1/5)$?
Step 2 Kangaroo / MATHCOUNTS	What is the product of $(1 - 1/s)$ for $s = 2, 3, 4, ..., 100$?
Step 3 MATHCOUNTS	What is the smallest number of different whole numbers (each at least 2) whose product of $(1 - 1/s)$ is exactly 1/10?
Step 4 AMC 8	Ana uses the numbers 2, 3, ..., 34 except 26, and one more number m bigger than 34. Her product is exactly 3/100. What is m?
Step 5 AMC 10	Using exactly 20 different whole numbers (each at least 2), what is the smallest possible value of the product?
Challenge AIME	What is the smallest number of different whole numbers (each at least 2) whose product of $(1 - 1/s)$ is exactly 7/200?

Ladder M — The Average Divisor Count

$\lfloor x \rfloor$ means x rounded down to a whole number. d(n) is the number of divisors of n. Let $f(n) = (\lfloor n/1 \rfloor + \lfloor n/2 \rfloor + \ldots + \lfloor n/n \rfloor) / n$.

Step 1 Kangaroo	What is d(12), the number of divisors of 12?
Step 2 Kangaroo / MATHCOUNTS	What is $\lfloor 6/1 \rfloor + \lfloor 6/2 \rfloor + \lfloor 6/3 \rfloor + \lfloor 6/4 \rfloor + \lfloor 6/5 \rfloor + \lfloor 6/6 \rfloor$?
Step 3 MATHCOUNTS	What is $d(1) + d(2) + \ldots + d(10)$?
Step 4 AMC 8	What is f(10)?
Step 5 AMC 10	What is the smallest n with $f(n) > 3$?
Challenge AIME	For how many n from 1 to 100 is $f(n + 1) < f(n)$?

Ladder N — Values That Share a Factor

$P(n) = n^2 + n + 1$, so $P(1) = 3$, $P(2) = 7$, $P(3) = 13$, $P(4) = 21$, ... A block $P(a + 1), P(a + 2), \ldots, P(a + 6)$ of six consecutive values is **friendly** if every value in it shares a common factor bigger than 1 with at least one other value in the block.

Step 1 Kangaroo	Among $P(1), \ldots, P(6) = 3, 7, 13, 21, 31, 43$, how many pairs share a common factor bigger than 1?
Step 2 Kangaroo / MATHCOUNTS	What is the greatest common divisor of P(10) and P(11)?
Step 3 MATHCOUNTS	What is the smallest $n \geq 1$ for which P(n) and $P(n + 2)$ share a factor bigger than 1?
Step 4 AMC 8	P(n) and $P(n + 2)$ share a factor bigger than 1 only when n leaves the same remainder r on division by 7. What is r?
Step 5 AMC 10	For how many n from 1 to 30 do P(n) and $P(n + 3)$ share a factor bigger than 1?
Challenge AIME	What is the smallest $a \geq 0$ for which the block $P(a + 1), \ldots, P(a + 6)$ is friendly?

Ladder O — Pick With Small Gaps

Ana writes a row of numbers, each $+1$ or -1. Ben then picks some of them, keeping their order. Between two numbers Ben picks, he may skip **at most one** number. He may start and stop anywhere. Ben's **score** is the size of the sum of his picks, ignoring the sign (a sum of -3 scores 3).

Step 1 Kangaroo	The row is $+1, -1, -1, +1, +1, -1$. What is Ben's best score?
Step 2 Kangaroo / MATHCOUNTS	The row is $+1, -1, -1, +1, +1, -1, -1, +1, +1, -1$. What is Ben's best score?
Step 3 MATHCOUNTS	Ana writes 6 numbers and wants Ben's best score to be as small as possible. What is the largest score Ben can be sure to reach, whatever Ana writes?
Step 4 AMC 8	The same with 10 numbers.
Step 5 AMC 10	The same with 14 numbers.
Challenge **AIME**	The same with 2026 numbers.

Ladder P — Rainbow Necklaces

m marbles sit in a circle, each painted one of n colours. The circle is a **rainbow necklace** if any $n + 1$ marbles in a row (going around the circle) include all n colours.

Step 1 Kangaroo	With 3 colours, for how many of the numbers $m = 3, 4, 5, \ldots, 10$ is there a rainbow necklace of m marbles?
Step 2 Kangaroo / MATHCOUNTS	With 3 colours, which number of marbles $m \geq 3$ is impossible?
Step 3 MATHCOUNTS	With 4 colours, what is the largest number of marbles that is impossible?
Step 4 AMC 8	With 4 colours, every colour must appear among any 5 marbles in a row. In a necklace of 11 marbles, at least how many times must each colour appear?
Challenge **AIME**	With 6 colours, for how many numbers $m \geq 6$ is a rainbow necklace of m marbles impossible?

Ladder Q — Flip the k-th Coin

A row of coins shows heads (H) or tails (T). While at least one coin shows H: if k coins show H, flip the k-th coin from the left. Count the steps until every coin shows T.

Step 1 Kangaroo	How many steps does the row H T H take?
Step 2 Kangaroo / MATHCOUNTS	How many steps does the row H H H take?
Step 3 MATHCOUNTS	Add up the steps for all four rows of 2 coins (HH, HT, TH, TT).
Step 4 AMC 8	Add up the steps for all eight rows of 3 coins.
Step 5 AMC 10	What is the average number of steps over all 16 rows of 4 coins?
Challenge **AIME**	What is the average number of steps over all 4096 rows of 12 coins?

Ladder R — Rooks and Empty Squares

An n × n board has n rooks, exactly one in each row and each column. An **empty square** is a k × k block of neighbouring cells with no rook in it.

Step 1 Kangaroo	On a 4 × 4 board the rooks in rows 1, 2, 3, 4 stand in columns 2, 4, 1, 3. What is the side of the largest empty square?
Step 2 Kangaroo / MATHCOUNTS	What is the largest k such that every 4 × 4 board has an empty k × k square?
Step 3 MATHCOUNTS	The same for a 5 × 5 board.
Step 4 AMC 8	The same for a 10 × 10 board.
Step 5 AMC 10	What is the smallest n such that every n × n board has an empty 3 × 3 square?
Challenge **AIME**	What is the largest k such that every 26 × 26 board has an empty k × k square?

Ladder S — Pairs With Different Sums

From the numbers 1, 2, ..., n, choose some pairs. No number may be in two pairs, all pair sums must be different, and every pair sum must be at most n.

Step 1 Kangaroo	With n = 8, what is the largest number of pairs?
Step 2 Kangaroo / MATHCOUNTS	With n = 13, what is the largest number of pairs?
Step 3 MATHCOUNTS	What is the smallest possible total of 10 different positive whole numbers?
Step 4 AMC 8	What is the largest possible total of 5 different whole numbers, each at most 20?
Step 5 AMC 10	Steps 3 and 4 give an inequality for x pairs. For n = 100, what is the largest x it allows?
Challenge **AIME**	With n = 2025, what is the largest number of pairs?

Ladder T — Triples Without Repeats

A triple (a, b, c) of whole numbers 0, 1, 2, ... with a + b + c = n. A set of such triples is **clean** if no two triples have the same first number, no two have the same second number, and no two have the same third number.

Step 1 Kangaroo	With n = 2, what is the largest clean set?
Step 2 Kangaroo / MATHCOUNTS	With n = 3, what is the largest clean set?
Step 3 MATHCOUNTS	What is the smallest possible total of 7 different whole numbers (0 allowed)?
Step 4 AMC 8	With n = 9, what is the largest clean set?
Step 5 AMC 10	Step 3 gives an inequality for a clean set of N triples. For n = 30, what is the largest N it allows?
Challenge **AIME**	With n = 2026, what is the largest clean set?

Ladder U — Doubling Prefix Sums

Write the numbers 1, 2, ..., n in some order. The order is **good** if for every k = 1, 2, ..., n, the number k divides 2 × (the sum of the first k numbers).

Step 1 Kangaroo	How many orders of 1, 2, 3 are good?
Step 2 Kangaroo / MATHCOUNTS	How many orders of 1, 2, 3, 4 are good?
Step 3 MATHCOUNTS	For n = 5, which numbers can be last in a good order? Give their sum.
Step 4 AMC 8	How many orders of 1, 2, ..., 6 are good?
Challenge **AIME**	Let G be the number of good orders of 1, 2, ..., 2026. What are the last three digits of G?

Ladder V — Same-Colour Triples

Colour each of the numbers 1, 2, ..., n red or blue. Count the ordered triples (x, y, z) of these numbers (repeats allowed) such that x, y, z all have the same colour and x + y + z is divisible by n.

Step 1 Kangaroo	n = 3 and all three numbers are red. How many triples are counted?
Step 2 Kangaroo / MATHCOUNTS	n = 4, with 1 and 2 red and 3 and 4 blue. How many triples are counted?
Step 3 MATHCOUNTS	Forget the colours: for n = 10, how many ordered triples (x, y, z) from 1 to 10 have x + y + z divisible by 10?
Step 4 AMC 8	n = 7, with 1, 2, 3 red and 4, 5, 6, 7 blue. How many triples are counted?
Step 5 AMC 10	n = 10. What is the smallest possible number of counted triples, over all colourings?
Challenge **AIME**	Exactly 2028 triples are counted. Find the sum of all n for which this is possible.

Ladder W — Lattice Friends

A lattice point has whole-number coordinates. Two lattice points A and B are **k-friends** if some lattice point C makes triangle ABC have area exactly k (k is a positive whole number). A **k-team** is a set of lattice points in which every two are k-friends.

Step 1 Kangaroo	What is the area of the triangle with corners (0, 0), (4, 0), (0, 3)?
Step 2 Kangaroo / MATHCOUNTS	What is the smallest whole number k ≥ 1 for which (0, 0) and (6, 9) are k-friends?
Step 3 MATHCOUNTS	How many points (u, v) with 1 ≤ u ≤ 12 and 1 ≤ v ≤ 12 are 6-friends of (0, 0)?
Step 4 AMC 8	What is the smallest k for which the 16 points (x, y) with 0 ≤ x ≤ 3 and 0 ≤ y ≤ 3 form a k-team?
Challenge **AIME**	What is the smallest k for which some k-team has more than 100 points?

Ladder X — Floors of Quotients

Write 1, 2, ..., n in some order $a_1, a_2, \ldots, a_n$. The **score** is $\lfloor a_1/1 \rfloor + \lfloor a_2/2 \rfloor + \ldots + \lfloor a_n/n \rfloor$, where $\lfloor x \rfloor$ means x rounded down.

Step 1 Kangaroo	What is the score of the order 1, 2, 3?
Step 2 Kangaroo / MATHCOUNTS	For $n = 3$, what is the smallest possible score?
Step 3 MATHCOUNTS	For $n = 4$, what is the smallest possible score?
Step 4 AMC 8	For $n = 8$, what is the smallest possible score?
Step 5 AMC 10	For $n = 7$, what is the score of the order 1, 3, 2, 7, 4, 5, 6?
Challenge **AIME**	For $n = 2026$, what is the smallest possible score?

Ladder Y — Sums That Divide the Total

Take a set of four different positive whole numbers. Count the pairs from the set whose sum divides the total of all four numbers.

Step 1 Kangaroo	For the set {1, 2, 3, 4}, how many pairs are counted?
Step 2 Kangaroo / MATHCOUNTS	What is the largest count any set can have?
Step 3 MATHCOUNTS	For the set {1, 5, 7, 11}, how many pairs are counted?
Step 4 AMC 8	What is the smallest possible total of a set with count 4?
Challenge **AIME**	How many sets with all four numbers at most 100 have count 4?

Ladder Z — Average of the Last Few

A robot records readings $r_0 = 0$ and $r_1 = 1$. On each later day $n = 2, 3, \ldots$, it chooses a number k with $1 \le k \le n$ and records r_n = the average of the last k readings $r_{n-1}, r_{n-2}, \ldots, r_{n-k}$.

Step 1 Kangaroo	On day 2 the robot chooses $k = 2$. What is r_2?
Step 2 Kangaroo / MATHCOUNTS	What is the sum of all possible values of r_2?
Step 3 MATHCOUNTS	What is the largest possible value of $r_3 - r_2$?
Step 4 AMC 8	What is the largest possible value of $r_5 - r_4$?
Challenge **AIME**	The largest possible value of $r_9 - r_8$ is p/q in lowest terms. What is $p + q$?

Ladder AA — The Inner Arc

In triangle ABC, I is the incentre (the centre of the circle touching all three sides).

Step 1 Kangaroo	A triangle has sides AB = 13, BC = 14, CA = 15. What is its area?
Step 2 Kangaroo / MATHCOUNTS	What is the radius of the circle inside the triangle of Step 1 that touches all three sides?
Step 3 MATHCOUNTS	In a triangle with $\angle A = 60°$, what is $\angle BIC$ in degrees?
Step 4 AMC 8	In the triangle of Step 1, what is AI^2?
Challenge AIME	Triangle ABC has AB = 10, BC = 21, CA = 17. A point P inside it satisfies $\angle PBA + \angle PCA = \angle PBC + \angle PCB$. What is the smallest possible value of $4 \times AP^2$?

Ladder AB — Powers Plus Fourth Powers

p is a prime and a is a positive whole number. We ask when $p^a + a^4$ is a perfect square.

Step 1 Kangaroo	What is $3^2 + 2^4$?
Step 2 Kangaroo / MATHCOUNTS	For how many a from 1 to 50 is $2^a + a^4$ a perfect square?
Step 3 MATHCOUNTS	$3^6 + 6^4$ is a perfect square. What is its square root?
Step 4 AMC 8	$3^9 + 9^4 = y^2$. What is y?
Challenge AIME	Find the sum of all a for which some prime p makes $p^a + a^4$ a perfect square.

Part 2 — Answer key

Ladder	1	2	3	4	5	Challenge
I	2	6	11	60		**1348**
J	2	0	2	2	18	**334**
K	8	6	24	14	7560	**7**
L	1/5	1/100	9	52	1/21	**28**
M	6	14	27	27/10	16	**43**
N	2	1	2	2	10	**195**
O	2	3	2	3	4	**507**
P	7	5	11	3		**10**
Q	4	3	6	24	5	**39**
R	2	1	2	3	10	**5**
S	3	5	55	90	39	**809**
T	2	3	21	7	21	**1351**
U	6	12	6	48		**648**
V	9	4	100	13	25	**216**
W	6	3	136	3		**1260**
X	3	2	3	4	3	**11**
Y	2	4	4	24		**12**
Z	1/2	3/2	1/4	3/16		**71**
AA	84	4	120	65		**85**
AB	25	0	45	162		**18**

Every answer is recomputed by the verification script that goes with the draft (see the draft's header).

Part 3 — Solutions and notes

Ladder I — Three Divisors Make a Total

> **Idea card (internal):** SL2022-N1 (write the three divisors as N/a, N/b, N/c; the target is N × (1/a + 1/b + 1/c), and a small N forces a large bracket, which leaves only a few cases).

- Step 1: With 12: the other two must add to 7, which gives {1, 6} or {3, 4}. Without 12: at most 6 + 4 + 3 = 13. So {1, 6, 12} and {3, 4, 12}.
- Step 2: 1 + 2 + 3 = 6. The numbers 1 to 5 have too few divisors, and 4 gives 1 + 2 + 4 = 7.
- Step 3 (the key idea): dividing by N gives 1/a + 1/b + 1/c = 1. Here a = 1 is impossible (the other two fractions would add to 0), so a ≥ 2. Then a = 2 forces 1/b + 1/c = 1/2, so b = 3, c = 6 (b = 4 needs c = 4). And a = 3 is too small: 1/3 + 1/4 + 1/5 < 1. So a + b + c = 2 + 3 + 6 = 11.
- Step 4: 100 = N(1/a + 1/b + 1/c). If N < 60 the bracket is more than 5/3. That forces a = 1 and b = 2, because a ≥ 2 gives at most 1/2 + 1/3 + 1/4 = 13/12. Then 1/c > 1/6, so c is 3, 4 or 5. None of these makes N a whole number (11N/6, 7N/4 and 17N/10 would have to equal 100). N = 60 works: 60 + 30 + 10 = 100.
- Challenge: If N < 1348, the bracket is more than 2026/1348 > 1.5, so a = 1 as in Step 4.
  - If b = 3, then c is 4 or 5, and N = 2026 × 12/19 or 2026 × 15/23, neither a whole number.
  - If b = 2, then (3c + 2)N = 4052c. Since 3c + 2 and c share at most a factor 2, 3c + 2 divides 8104 = 8 × 1013. With c > 2, this leaves c = 337 (N = 1348) or c = 1350 (N = 1350).
  - So the smallest is N = 1348: 1348 + 674 + 4 = 2026.
  - Computer check: every N up to 1348.

Explore — *Status: not searched in the literature yet.* Make a table of the smallest N for T = 10, 11, ..., 60. How often is it 3T/5, as for T = 100 (60 + 30 + 10, the divisors N, N/2, N/6)? What happens when T is a prime?

Research field: elementary number theory; Egyptian fractions (sums of unit fractions).

Ladder J — Root or Add Three

> **Idea card (internal):** SL2017-N1 (look at remainders when dividing by 3: numbers that leave remainder 2 are never perfect squares, so once there the sequence grows forever; multiples of 3 keep coming down and end in a cycle).

- Step 1: 7, 10, 13, 16, 4, 2.
- Step 2: 5, 8, 11, ... all leave remainder 2 when divided by 3, and (Step 3) no square does. None.
- Step 3: squares of 3k, 3k + 1, 3k + 2 leave remainders 0, 1, 1. Two remainders: 0 and 1.
- Step 4: 2026, 2029, ..., 2116 = 46^2, then 46, 49, 7, 10, 13, 16, 4, 2, 5, 8, ... After 2 every number leaves remainder 2, so the numbers only grow. The smallest is 2.
- Step 5: 30, 33, 36, 6, 9, 3, 6, 9, 3, ... The cycle is 3, 6, 9: sum 18.
- Challenge (the key idea: sort the starting numbers by remainder):
  - Remainder 2: never a square, so the numbers grow forever and nothing repeats.
  - Remainder 0: every number stays a multiple of 3 (a square that is a multiple of 3 has a root that is a multiple of 3). Above 9, adding 3 soon reaches a square $(3t)^2$ whose root 3t is much smaller, so the numbers come down until they are at most 9, and then 3, 6, 9 repeats. All 333 multiples of 3 repeat.
  - Remainder 1: adding 3 keeps remainder 1 until a square is reached; its root is smaller and is not a multiple of 3. If the root leaves remainder 2, the sequence grows forever. If it leaves remainder 1, the

same thing happens again with a smaller number. So the numbers keep coming down until a root with remainder 2 appears (for small numbers: $4 \to 2$, $25 \to 5$, $64 \to 8$). The only exception is the start 1, because $\sqrt{1} = 1$ repeats at once.
    - Total: $333 + 1 = 334$. Computer check: all 1000 starts, followed until a repeat or until a number with remainder 2 appears (from then on nothing can repeat).

Explore — *Status: not searched in the literature yet.* Replace "add 3" by "add 5" or "add 7". Which starting numbers now lead to a repeat? (Look at the remainders a square can leave when divided by 5 or 7.)

Research field: discrete dynamical systems; modular arithmetic.

Ladder K — Divisor Counts That Double

> **Idea card (internal):** SL2011-N1 (a number has a power-of-2 count of divisors exactly when it is a product of different "blocks" p, p^2, p^4, p^8, ...; each block doubles the count, so the smallest such number uses the smallest blocks).

- Step 1: $24 = 2^3 \times 3$ has $(3 + 1)(1 + 1) = 8$ divisors.
- Step 2: 1, 2, 3, 4, 5 have 1, 2, 2, 3, 2 divisors; 6 has 1, 2, 3, 6.
- Step 3: $24 = 2 \times 3 \times 4$, and no smaller number has 8 divisors.
- Step 4: the primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (10), the squares 4, 9, 25 (3) and $16 = 2^4$ (1): 14.
- Step 5 (the key idea): d(n) is a product of (exponent + 1). It is a power of 2 exactly when every exponent + 1 is a power of 2, that is, every exponent is 1, 3, 7, 15, ... Such a prime power splits into different blocks ($2^7 = 2 \times 2^2 \times 2^4$), and each block doubles d(n). So a number with 2^k divisors is a product of k different blocks, and the smallest one takes the k smallest blocks. For $64 = 2^6$: $2 \times 3 \times 4 \times 5 \times 7 \times 9 = 7560$.
- Challenge: $1024 = 2^{10}$, so N is the product of the 10 smallest blocks: 2, 3, 4, 5, 7, 9, 11, 13, 16, 17. $N = 2^7 \times 3^3 \times 5 \times 7 \times 11 \times 13 \times 17$ uses 7 different primes. Computer check: a separate search over all exponent patterns gives the same N = 294 053 760.

Explore — *Status: known (smallest numbers with a given number of divisors are listed in the OEIS).* What is the smallest number with exactly 12 divisors? With $2^{10} \times 3$ divisors? Is the "blocks" idea still enough when the count is not a power of 2?

Research field: multiplicative number theory (the divisor function).

Ladder L — The Shrinking Product

> **Idea card (internal):** SL2010-N1 (sort the numbers: the i-th smallest is at least i + 1, so the product is at least (1/2)(2/3)...(n/(n + 1)) = 1/(n + 1); then build an example with one carefully chosen gap).

- Steps 1–2: (1/2)(2/3)(3/4)(4/5) = 1/5: everything cancels except the first top and last bottom. Up to 100: 1/100.
- Step 3 (the key idea): sort the numbers $s_1 < s_2 < \ldots < s_n$. Then $s_i \geq i + 1$, so each factor is at least $1 - 1/(i + 1)$, and the product is at least $1/(n + 1)$. For 1/10 we need $n + 1 \geq 10$. The numbers 2, ..., 10 give exactly 1/10, so the answer is 9.
- Step 4: 2, ..., 34 give 1/34. Dropping 26 multiplies by 26/25, and adding m multiplies by $(m - 1)/m$. So $(1/34)(26/25)(m - 1)/m = 3/100$, which gives $(m - 1)/m = 51/52$ and $m = 52$.
- Step 5: by Step 3 the product is at least 1/21, and 2, ..., 21 give exactly 1/21.
- Challenge: with n numbers the product is at least $1/(n + 1)$. With 27 numbers it is at least 1/28, which is more than 7/200 (because $200 > 7 \times 28 = 196$), so we need at least 28. Example with 28 numbers:

2, ..., 29 without 21, plus 30. Product $(1/29)(21/20)(29/30) = 7/200$.

Explore — *Status: not searched in the literature yet.* For the target 9/250 the bound from Step 3 says "at least 27". Can 27 numbers do it, or are 28 needed?

Research field: elementary number theory; telescoping products.

Ladder M — The Average Divisor Count

> **Idea card (internal):** SL2006-N3 ($\lfloor n/k \rfloor - \lfloor (n-1)/k \rfloor$ is 1 exactly when k divides n, so $\lfloor n/1 \rfloor + \ldots + \lfloor n/n \rfloor$ is $d(1) + \ldots + d(n)$; the function is really an average of divisor counts).

- Step 1: 1, 2, 3, 4, 6, 12.
- Step 2: $6 + 3 + 2 + 1 + 1 + 1 = 14$. (Also $d(1) + \ldots + d(6) = 1 + 2 + 2 + 3 + 2 + 4 = 14$.)
- Step 3: $1 + 2 + 2 + 3 + 2 + 4 + 2 + 4 + 3 + 4 = 27$.
- Step 4 (the key idea): $\lfloor n/k \rfloor$ counts the multiples of k up to n. Adding over k counts every pair (k, m) with k dividing $m \le n$, which is $d(1) + \ldots + d(n)$. So f(n) is the **average** of $d(1), \ldots, d(n)$, and $f(10) = 27/10$.
- Step 5: $d(11), \ldots, d(16)$ are 2, 6, 2, 4, 4, 5, so the sums are 45 at $n = 15$ ($f = 3$ exactly) and 50 at $n = 16$ ($f = 3.125$).
- Challenge: adding a new value to an average lowers it exactly when the new value is below the average, so $f(n+1) < f(n)$ exactly when $d(n+1) < f(n)$. The average is at most about 4.8 for $n \le 100$. So the drops come from $n + 1$ a prime from 7 to 101 ($d = 2$; at 2, 3 and 5 the average is still too small: 23 cases), a prime squared once the average is above 3 ($d = 3$: 25 and 49), and $d = 4$ (numbers p^3 or $p \times q$ once the average is above 4: 18 cases). Total $23 + 2 + 18 = 43$. Computer check: f computed exactly for every $n \le 100$.

Explore — *Status: the growth of the average is known (Dirichlet's divisor problem, see below).* Up to 1000 the computer finds 518 drops and only one n with $f(n+1) = f(n)$. Which n is it? Do drops and rises come out about even forever?

Research field: analytic number theory: the average of d(n) is about ln n + 0.15, and how precise that estimate can be is Dirichlet's divisor problem, still open.

Ladder N — Values That Share a Factor

> **Idea card (internal):** SL2016-N3 (for $P(n) = n^2 + n + 1$, neighbouring values never share a factor, values two apart can only share 7, values three apart can only share 3; a block where every value has a partner needs several congruences at once, solved by the Chinese remainder theorem).

- Step 1: $21 = 3 \times 7$ shares 3 with 3 and 7 with 7. Two pairs.
- Step 2: $P(10) = 111 = 3 \times 37$ and $P(11) = 133 = 7 \times 19$: gcd 1. In general a common factor of P(n) and P(n + 1) divides their difference $2(n + 1)$; it is odd, so it divides $n + 1$ and then $P(n) - n(n + 1) = 1$.
- Step 3: $P(2) = 7$ and $P(4) = 21$.
- Step 4: a common factor d of P(n) and P(n + 2) divides the difference $4n + 6$, so $2n \equiv -3 \pmod{d}$, and $4P(n) = (2n + 1)^2 + 3 \equiv 4 + 3 = 7$. So $d = 7$, which needs $n \equiv 2 \pmod{7}$.
- Step 5: the same trick for $P(n + 3) - P(n) = 6(n + 2)$ gives $d = 3$, which divides P(n) exactly when $n \equiv 1 \pmod{3}$: 10 values.

- Challenge: values 1 apart never share; 2 apart share only 7; 3 apart only 3; 4 apart only 19 (same trick); 5 apart only 7. Each of the six values needs a partner, and these rules allow very few patterns. The smallest block is a = 195: P(196) and P(199) share 3, P(198) and P(200) share 7, P(197) and P(201) share 19. Computer check: every a < 195 fails. No block of 2 to 5 values is friendly for any a below 100 000.

Explore — *Status: not searched in the literature yet.* Try $Q(n) = n^2 + 1$ instead. Which pairs of values can share a factor? Is there a friendly block of six?

Research field: elementary number theory (the Chinese remainder theorem; prime factors of quadratic polynomials).

Ladder O — Pick With Small Gaps

> **Idea card (internal):** SL2022-C1 (take the more common sign; walk along taking every number of that sign and, in each run of the other sign, skipping every other number; at most half of the other sign is ever taken).

- Step 1: the +1s are in places 1, 4, 5. Using all three means also picking place 2 or 3 (−1): total 2. The −1s (places 2, 3, 6) need place 4 or 5 in between: total −2. Picking 4 and 5 alone also gives 2. Best: 2.
- Step 2: pick places 2, 3, 4, 6, 7: $-1 - 1 + 1 - 1 - 1 = -3$. Computer check: nothing scores 4.
- Steps 3–5: computer check of every row (64, 1024 and 16 384 rows).
- Challenge (the key idea): say +1 appears $P \geq 1013$ times. Ben picks every +1. Between two +1s there is a run of −1s; skipping every other one, he picks about half of each run (none at the two ends of the row). So he picks at most half of the $2026 - P$ minus ones, and his sum is at least $P - (2026 - P)/2 = (3P - 2026)/2 \geq 506.5$, so at least 507. Ana's row +1, −1, −1, +1, +1, −1, −1, +1, +1, ... allows only 507 (computer check). The general answer is $\lfloor (n + 1)/4 \rfloor + 1$, checked for every row of up to 18 numbers.

Explore — *Status: not searched in the literature yet.* Allow Ben to skip at most **two** numbers in a row. What can he always reach with 12 numbers? With 2026?

Research field: combinatorics on words; discrepancy theory.

Ladder P — Rainbow Necklaces

> **Idea card (internal):** SL2021-C2 (each colour must appear in every window of n + 1 marbles; counting how often each colour appears gives the impossible lengths, and two kinds of blocks build all the others).

- Steps 1–2: computer search. Only 5 fails; for example 6 marbles: 1 2 3 1 2 3.
- Step 3: computer search: 6, 7 and 11 fail; 11 is the largest.
- Step 4 (the key idea): going around the circle, the gaps between two marbles of the same colour hold at most 4 other marbles, so each colour appears at least $\lceil 11/5 \rceil = 3$ times. That needs $4 \times 3 = 12$ marbles, more than 11. So 11 is impossible.
- Challenge: the same count shows m is impossible whenever $6 \times \lceil m/7 \rceil > m$. In every other case m can be written as 6a + 7b, and gluing a blocks "1 2 3 4 5 6" and b blocks "1 1 2 3 4 5 6" gives a rainbow necklace. The numbers $m \geq 6$ that are not 6a + 7b are 8, 9, 10, 11, 15, 16, 17, 22, 23, 29: ten numbers ($29 = 6 \times 7 - 6 - 7$ is the largest, as in the coin problem). Computer check: for 3, 4, 5 colours the counting rule matches a full search, and every glued necklace up to 200 marbles is checked.

Explore — *Status: not searched in the literature yet.* Change the rule to "any n + 2 marbles in a row include all n colours". Which numbers of marbles become impossible?

Research field: combinatorics on words (necklaces); the Frobenius coin problem.

Ladder Q — Flip the k-th Coin

> **Idea card (internal):** SL2019-C3 (a row ending in T behaves like the shorter row; a row ending in H is a flipped, reversed shorter row plus n extra steps; this recursion gives the average).

- Step 1: H T H (2 heads, flip coin 2) → H H H → H H T → H T T → T T T: 4 steps.
- Step 2: H H H → H H T → H T T → T T T: 3 steps.
- Steps 3–4: HH 2, HT 1, TH 3, TT 0: total 6. For 3 coins the eight rows take 0, 1, 2, 3, 3, 4, 5, 6 steps: total 24.
- Step 5 (the key idea): if the last coin is T it is never flipped (k is at most the number of other coins), so the row takes as long as the shorter row. If the last coin is H, the row takes n steps more than the shorter row with every coin flipped and read backwards. So the total for n coins is $2 \times$ (total for $n - 1$) $+ n \times 2^{n-1}$, and the average grows by $n/2$ each time: 1/2, 3/2, 3, 5, ... For 4 coins: 5.
- Challenge: the average is $1/2 + 2/2 + \ldots + n/2 = n(n + 1)/4$. For 12 coins: 39. Computer check: all 4096 rows, and both rules above for every row of up to 10 coins.

Explore — *Status: not searched in the literature yet.* Which row of n coins takes the most steps? How many steps does it take?

Research field: combinatorics of processes; recursive structures (the process is closely related to Gray codes).

Ladder R — Rooks and Empty Squares

> **Idea card (internal):** SL2014-C3 (if $n > \ell^2$, look at ℓ rows next to each other that include the rook in the first column: cutting the board into ℓ squares of side ℓ leaves one without a rook; for $n \le \ell^2$ a spread-out board avoids empty $\ell \times \ell$ squares).

- Step 1: rows 2–3 hold rooks in columns 4 and 1, so the block rows 2–3, columns 2–3 is empty. A 3×3 block covers 3 rows, whose 3 rooks sit in only 4 columns, and the computer confirms none of the four 3×3 blocks is empty. So 2.
- Step 2: the board with rooks in columns 1, 3, 2, 4 has no empty 2×2 square.
- Steps 3–5: computer search over every board (with pruning). The answers follow $\lfloor \sqrt{(n - 1)} \rfloor$.
- Challenge (the key idea): take $\ell = 5$, so $n = 26 > 25$. Take 5 neighbouring rows that include the row whose rook is in column 1. They hold 5 rooks. Drop the first column (and its rook): the remaining 25 columns of these rows form 5 squares of side 5 holding at most 4 rooks, so one is empty. For $k = 6$, the board with rooks in columns 1, 7, 12, 17, 22, 2, 8, 13, 18, 23, 3, 9, 14, 19, 24, 4, 10, 15, 20, 25, 5, 11, 16, 21, 26, 6 has no empty 6×6 square. So the answer is 5. Computer check: a full search shows every 26×26 board has an empty 5×5, and the board above is checked.

Explore — *Status: not searched in the literature yet.* How many 5×5 boards have no empty 2×2 square? How many 10×10 boards have no empty 4×4 square?

Research field: extremal combinatorics (permutation patterns; "empty boxes" in point sets).

Ladder S — Pairs With Different Sums

> **Idea card (internal):** SL2012-C2 (the 2x chosen numbers add up to at least $1 + 2 + \ldots + 2x$, while x different sums of at most n add up to at most $n + (n - 1) + \ldots$; comparing gives $x \le (2n - 1)/5$, and a two-part pattern reaches it).

- Step 1: (1, 5), (2, 6), (3, 4) with sums 6, 8, 7. Computer check: 4 pairs are impossible.
- Step 2: (1, 8), (2, 9), (3, 10), (4, 6), (5, 7) with sums 9, 11, 13, 10, 12.
- Steps 3–4: $1 + \ldots + 10 = 55$ and $20 + 19 + 18 + 17 + 16 = 90$.
- Step 5 (the key idea): x pairs use 2x different numbers, total at least $x(2x + 1)$. Their x different sums are at most n, total at most $n + (n - 1) + \ldots + (n - x + 1) = x(2n - x + 1)/2$. So $2(2x + 1) \le 2n - x + 1$, i.e. $x \le (2n - 1)/5$. For $n = 100$: $x \le 39.8$, so 39.
- Challenge: the bound gives $x \le 4049/5$, so at most 809. With $k = 404$ ($809 = 2k + 1$), the pairs $(k - i, 2k + 3 + 2i)$ for $i = 0..k - 1$ and $(2k + 1 - i, 2k + 2 + 2i)$ for $i = 0..k$ use the numbers 1 to 1618 and have the sums 1215 to 2023, all different and at most 2025. Computer check: the 809 pairs, and a full search for every n up to 15.

Explore — *Status: not searched in the literature yet.* Ask for triples instead of pairs: different triple sums, each at most n. Find the best number for $n = 20$ by hand or by computer, and guess a rule.

Research field: additive combinatorics (Skolem sequences and Langford pairings).

Ladder T — Triples Without Repeats

> **Idea card (internal):** SL2009-C2 (with N triples, the first entries are N different numbers, so they add up to at least $0 + 1 + \ldots + (N - 1)$; the same for the other two positions; adding gives $3N(N - 1)/2 \le Nn$).

- Step 1: (0, 0, 2) and (1, 1, 0). Computer check: no three triples work.
- Step 2: (0, 1, 2), (1, 2, 0), (2, 0, 1).
- Step 3: $0 + 1 + \ldots + 6 = 21$.
- Step 4: computer search: 7, for example (0, 3, 6), (1, 4, 4), (2, 5, 2), (3, 6, 0), (4, 0, 5), (5, 1, 3), (6, 2, 1).
- Step 5 (the key idea): add up all 3N numbers of a clean set of N triples. It is Nn, because each triple adds to n. It is also at least $3 \times (0 + 1 + \ldots + (N - 1)) = 3N(N - 1)/2$. So $N \le 2n/3 + 1$. For $n = 30$: $N \le 21$.
- Challenge: $N \le 2 \times 2026/3 + 1 = 1351.67$, so at most 1351. Example ($2026 = 3 \times 675 + 1$, $m = 675$): the triples $(i, m - 1 + i, \ldots)$ for $i = 0..m + 1$ and $(m + 2 + j, j, \ldots)$ for $j = 0..m - 2$, the third entry making the sum 2026: 1351 triples, all first, second and third entries different. Computer check: the 1351 triples, and a full search for n up to 12.

Explore — *Status: not searched in the literature yet.* Use quadruples (a, b, c, d) with $a + b + c + d = n$. What bound does the same trick give? Is it reached for $n = 10$?

Research field: combinatorial designs (Latin squares and transversals).

Ladder U — Doubling Prefix Sums

> **Idea card (internal):** SL2008-C2 (the rule for $k = n - 1$ forces the last number to be 1, $(n + 1)/2$ or n, and $k = n - 2$ rules out the middle one; each choice leaves a smaller good order, so the count doubles).

- Step 1: $k = 1$, 2 always work ($2 \times$ sum is even); $k = 3$: $2 \times 6 = 12$. All 6.
- Step 2: computer list: 12 orders; every one ends in 1 or 4.
- Step 3 (the key idea): for $k = n - 1$, $2 \times$ (sum of all but the last) $= n(n + 1) - 2 \times$ last must be divisible by $n - 1$. Since $n(n + 1) = (n - 1)(n + 2) + 2$, this means $n - 1$ divides $2 \times$ last $- 2$, so the

last number is 1, (n + 1)/2 or n. For n = 5, the middle choice 3 fails the k = 3 test, leaving 1 and 5: sum 6.

- Step 4: last = n leaves a good order of 1..n − 1; last = 1 leaves a good order of 2..n, which is a good order of 1..n − 1 shifted up by one. So the count doubles each time from n = 3 on: 6, 12, 24, 48.
- Challenge: $G = 6 \times 2^{2^{0}2^{3}} = 3 \times 2^{2^{0}2^{4}}$. Its last three digits are 648. Computer check: the count $3 \times 2^{n-2}$ for every n from 3 to 13, and $3 \times 2^{2^{0}2^{4}}$ mod 1000.

Explore — *Status: not searched in the literature yet.* Change "2 × sum" to "sum". The computer finds good orders only for n = 1 and n = 3. Why? What about "3 × sum"?

Research field: enumerative combinatorics (permutations with divisibility conditions).

Ladder V — Same-Colour Triples

> **Idea card (internal):** SL2007-C3 (count the mixed triples through their red-then-blue pairs: each mixed triple has exactly one, and each pair lies in exactly 3 triples; so the same-colour count depends only on how many numbers are red).

- Step 1: choose x and y freely (9 ways); then z is forced.
- Step 2: red triples from {1, 2} have sums 3 to 6, so the sum must be 4: (1, 1, 2) in 3 orders. Blue triples from {3, 4} have sums 9 to 12, so the sum must be 12: (4, 4, 4). Total 4.
- Step 3: z is forced by x and y: 10 × 10 = 100.
- Step 4 (the key idea): of the n^2 divisible triples, a mixed one (two of one colour, one of the other) contains exactly one "red then blue" pair among (x, y), (y, z), (z, x). Each red-then-blue pair (x, y) lies in exactly 3 divisible triples: (x, y, ·), (·, x, y), (y, ·, x). So there are 3rb mixed triples and $n^2 - 3rb = r^2 - rb + b^2$ same-colour ones (r red, b blue). For r = 3, b = 4: 9 − 12 + 16 = 13.
- Step 5: $r^2 - r(10 - r) + (10 - r)^2$ is smallest at r = 5: 25.
- Challenge: solve $r^2 - rb + b^2 = 2028$, i.e. $(2r - b)^2 + 3b^2 = 8112$. The solutions are {2, 46}, {26, 52}, {44, 46}, so n = 48, 78 or 90. Sum 216. Computer check: the formula against every colouring for n ≤ 7, and every n < 3000.

Explore — *Status: not searched in the literature yet.* Use three colours. Does the number of same-colour triples still depend only on how many numbers have each colour?

Research field: additive combinatorics (Schur-type counting problems).

Ladder W — Lattice Friends

> **Idea card (internal):** SL2008-C3 (a point (u, v) is a k-friend of the origin exactly when gcd(u, v) divides 2k; if some m does not divide 2k, two points with the same remainders mod m are not friends, so a team has at most m^2 points).

- Step 1: 4 × 3 / 2 = 6.
- Step 2 (the key idea): the triangle (0, 0), (u, v), (x, y) has area $|uy - vx|/2$, and uy − vx can be exactly the multiples of gcd(u, v). So (0, 0) and (u, v) are k-friends exactly when gcd(u, v) divides 2k. Here gcd(6, 9) = 3 must divide 2k: k = 3.
- Step 3: count the (u, v) with gcd(u, v) dividing 12: 136 (computer count; only the pairs with gcd 5, 7, 8, 9, 10, 11 fail).
- Step 4: the differences between the points are at most 3, so their gcd is 1, 2 or 3; 2k must be divisible by 2 and 3, so k = 3.

- Challenge: if some $m \le 10$ does not divide $2k$, two points of a team cannot have the same remainders mod m in both coordinates (their difference would have gcd divisible by m), so a team has at most $m^2 \le 100$ points. So $2k$ must be divisible by $1, 2, \ldots, 10$, that is by 2520: $k \ge 1260$. For $k = 1260$ the 121 points (x, y) with $0 \le x, y \le 10$ form a team (every gcd is at most 10). Computer check: the gcd rule against a direct search, the 121-point team, and every k below 1260.

Explore — *Status: not searched in the literature yet.* What is the largest 1-team? (Hint: think about remainders mod 3 and mod 4.)

Research field: geometry of numbers (lattice points, Pick's theorem).

Ladder X — Floors of Quotients

> **Idea card (internal):** SL2021-A3 (cut 1..n into blocks from 2^j to $2^{j+1} - 1$; inside each block put the largest number first, which costs exactly 1 per block, and an induction shows no order does better).

- Step 1: $1 + 1 + 1 = 3$.
- Step 2: 1, 3, 2 scores $1 + 1 + 0 = 2$. Every order scores at least 2: if 3 is in place 1 it alone scores 3; otherwise the number in place 1 scores at least 1 and 3 (in place 2 or 3) scores 1 more.
- Steps 3–4: computer search over all orders.
- Step 5 (the key idea): blocks $\{1\}$, $\{2, 3\}$, $\{4, 5, 6, 7\}$. In each block the largest number goes first and scores 1 (it is less than twice its place); every other number $i - 1$ sits in place i and scores 0. One point per block: 3.
- Challenge: the blocks 1, 2–3, 4–7, ..., 1024–2026 are 11 blocks, so the score 11 is possible. The idea card's induction shows that 2^k different positive numbers in any order score at least $k + 1$. The first $1024 = 2^{10}$ places already hold 1024 different numbers, so every order scores at least 11. Computer check: every order for $n \le 9$ (the answers are $\lfloor \log_2 n \rfloor + 1$), and the score of the 2026 example.

Explore — *Status: not searched in the literature yet.* Use $\lceil a_i / i \rceil$ (round up) instead. What is the smallest score for $n = 2026$?

Research field: combinatorial optimisation on permutations.

Ladder Y — Sums That Divide the Total

> **Idea card (internal):** SL2011-A1 (a pair sum divides the total exactly when it divides the other pair's sum; the two pairs with the largest number are too big, so at most 4 pairs work, and equality forces two families).

- Step 1: total 10: $1 + 4 = 5$ and $2 + 3 = 5$ divide 10; 3, 4, 6, 7 do not.
- Step 2 (the key idea): with $a < b < c < d$, a pair sum divides the total exactly when it divides the other two numbers' sum. The pairs $b + d$ and $c + d$ are bigger than their partners $a + c$ and $a + b$, so they cannot divide them. At most 4 pairs.
- Step 3: total 24: 6, 8, 12, 12, 16, 18; the sums 6, 8, 12, 12 divide 24.
- Step 4: count 4 forces $a + d = b + c$ and the remaining divisibilities, which leave only the families $\{k, 5k, 7k, 11k\}$ and $\{k, 11k, 19k, 29k\}$. The smallest total is $1 + 5 + 7 + 11 = 24$.
- Challenge: $11k \le 100$ gives $k = 1..9$, and $29k \le 100$ gives $k = 1..3$: $9 + 3 = 12$ sets. Computer check: all 3 921 225 sets of four numbers up to 100.

Explore — *Status: not searched in the literature yet.* Use sets of five numbers. What is the largest count now, and which sets reach it?

Research field: elementary number theory (divisibility).

Ladder Z — Average of the Last Few

> **Idea card (internal):** SL2018-A4 (the new reading and the one before it both lie between the smallest and largest possible averages; that spread shrinks by the factor $(n-1)/n$ each day, which bounds the jump).

- Steps 1–2: $k = 2$ gives $(1 + 0)/2 = 1/2$; $k = 1$ gives 1. Sum $3/2$.
- Step 3: $k = 2$ on day 2 ($r_2 = 1/2$) and $k = 2$ on day 3 ($r_3 = 3/4$): $1/4$. Computer check: all choices.
- Step 4: keep $k = 1$ on days 2 and 3 ($r_2 = r_3 = 1$), then $k = 4$ twice: $r_4 = 3/4$ and $r_5 = (3/4 + 3)/4 = 15/16$. The jump is $3/16$.
- Challenge (the key idea): the readings stay between the smallest and largest averages the robot could take, and that spread shrinks by $(n-1)/n$ each day. This gives $r_N - r_{(N-1)} \le (N-2)/(N-1)^2$, and the pattern of Step 4 reaches it: $k = 1$ until day $N - 2$, then $k = N - 1$ twice. For $N = 9$: $r_8 = 7/8$, $r_9 = 63/64$, jump $7/64$, and $7 + 64 = 71$. Computer check: every possible choice of k ($2 \times 3 \times \ldots \times 9 = 362\,880$ sequences).

Explore — *Status: not searched in the literature yet.* What is the largest possible value of $r_9 - r_7$? The smallest possible value of $r_9 - r_8$?

Research field: analysis of averaging processes (Cesàro means).

Ladder AA — The Inner Arc

> **Idea card (internal):** SL2006-G1 (the angle condition means $\angle BPC = 90° + A/2 = \angle BIC$, so P lies on the circle through B, I, C; its centre is on line AI, and the triangle inequality gives $AP \ge AI$).

- Step 1: Heron: $s = 21$, area $= \sqrt{(21 \times 8 \times 7 \times 6)} = 84$.
- Step 2: area $= r \times s$, so $r = 84/21 = 4$.
- Step 3: $\angle BIC = 180° - (B + C)/2 = 90° + A/2 = 120°$.
- Step 4: the incircle touches AB at distance $s - BC = 7$ from A, and I is $r = 4$ away from that point at a right angle: $AI^2 = 7^2 + 4^2 = 65$.
- Challenge (the key idea): the four angles add up to $B + C$, so the condition says $\angle PBC + \angle PCB = (B + C)/2$, that is $\angle BPC = 90° + A/2 = \angle BIC$. So P lies on the circle through B, I, C. Its centre M is where line AI meets the circle around ABC again, and $AP + PM \ge AM = AI + IM$ with $PM = IM$, so $AP \ge AI$, with equality at $P = I$. Here $s = 24$, area 84, $r = 7/2$, $s - BC = 3$, so $AI^2 = 9 + 49/4 = 85/4$ and $4 \times AP^2 \ge 85$. Computer check: P moved along that arc, 200 000 positions.

Explore — *Status: known (a classic lemma).* Show that M, the centre of the circle through B, I, C, is also the centre of a circle through the excentre opposite A. How far is that excentre from A in the triangle of Step 1?

Research field: Euclidean geometry (the incentre–excentre lemma).

Ladder AB — Powers Plus Fourth Powers

> **Idea card (internal):** SL2023-N2 ($p^a = y^2 - a^4 = (y - a^2)(y + a^2)$, so both factors are powers of p; their difference $2a^2$ is small compared with the larger one, which leaves only a few cases).

- Step 1: $9 + 16 = 25 = 5^2$.

- Step 2: computer check: none. (The Challenge argument below explains why $p = 2$ never works.)
- Step 3: $729 + 1296 = 2025 = 45^2$.
- Step 4 (the key idea): $y^2 - 9^4 = 3^9$, so $(y - 81)(y + 81) = 3^9$, and both factors are powers of 3 that differ by 162: 81×243. So $y = 162$.
- Challenge: in general $y - a^2 = p^i$ and $y + a^2 = p^j$ with $i < j$ and $i + j = a$, so $p^j - p^i = 2a^2$. Since $j > a/2$, this gives $p \hat{} (a/2) < 4a^2$, which allows only $p < 16$ and small a. Checking those cases leaves $(p, a) = (3, 1), (3, 2), (3, 6), (3, 9)$: $3 + 1 = 2^2$, $9 + 16 = 5^2$, $729 + 1296 = 45^2$, $19683 + 6561 = 162^2$. Sum of a: $1 + 2 + 6 + 9 = 18$. Computer check: every prime $p < 100$ and $a < 300$, which covers all cases allowed by the bound.

Explore — *Status: not searched in the literature yet.* Replace a^4 by a^2 or a^6. For which (p, a) is $p^a + a^2$ a perfect square?

Research field: exponential Diophantine equations.

Assessment

- **Size:** 20 ladders, 92 steps + 20 Challenges = 112 problems. 7 number theory, 9 combinatorics, 3 algebra, 1 geometry. Geometry is thin because most geometry ideas only give proofs, not numbers.
- **Checks:** every answer is recomputed by `verify_ladders_v0_3.py`. Where a Challenge is too big for brute force (O, R, S, T, U, W, X), the script checks an explicit example plus the counting bound, and compares the general formula with a full search on small cases.
- **Solutions that lean on the idea card:** the lower bounds in X (no order scores below 11) and the two families in Y are stated, not fully proved here; the computer checks cover them (X: all orders up to n = 9; Y: all sets up to 100).
- **Closest to the original problem** (same question, our numbers and story): S, T, U, V, Z. They need the same key idea as the original, so a reviewer should judge whether the change is large enough (rule 2.4).
- **Hardest by hand:** N (needs four congruences at once) and M (a careful count up to 100). Both may be above AIME level; consider them for the "Explore" slot instead if they test badly.
- **Not done yet:** independent solver pass, human review, web search for each Challenge's twist, famous open problems, and the "after the ladder" extras (similar olympiad problems, research, proof problems).